Enterohemorrhagic *Escherichia coli* (EHEC) O157:H7 is a bacterium responsible for causing foodborne illnesses. EHEC virulence, or the ability to cause disease, is acquired through quorum sensing, a form of bacterial cell-cell communication dependent on population density. EHEC O157:H7 bacteria monitor population density by detecting changes in the concentration of autoinducer-3 (AI-3), an extracellular signaling molecule secreted by many bacterial species. AI-3 generally acts on neighboring bacteria to activate the genes on the enterocyte effacement (*LEE*) locus. These genes code for the type III secretion system (T3SS) as well as several secreted proteins (eg, EspA, EspB, EspD, Tir), which promote entry of Shiga-like toxins secreted by EHEC into host intestinal epithelial cells. This ultimately leads to cell death, lesions, and symptoms such as bloody diarrhea and abdominal cramps.

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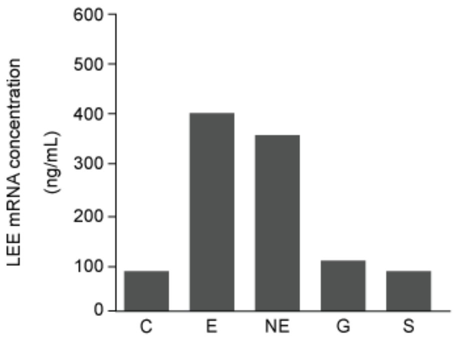


Figure 1 Concentration of *LEE* mRNA in mutant LuxS$^-$ bacteria supplemented with purified peptide hormones (Note: C = control.)

A western blot was performed to compare the presence and concentration of type III secretion proteins from wild-type (WT), LuxS$^-$, and LuxS$^-$ cultured in preconditioned media (PCM). PCM was generated by culturing uninfected intestinal epithelial cells in growth media for 24 hours. After 24 hours, the cells were removed and the PCM was used to supplement the growth of LuxS$^-$ bacteria.

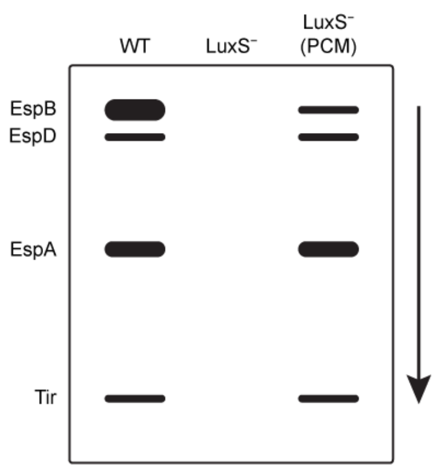


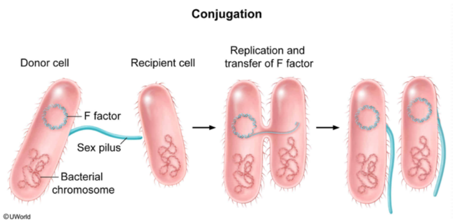
Figure 2 Western blot results of type III secretion proteins isolated from WT, mutant LuxS⁻, and LuxS⁻ in PCM

E. coli K-12, a normal bacterial strain of the human microbiota, is able to cause disease by inducing enterocyte lesions after it is grown in media with EHEC bacteria that contain the F factor plasmid. What process most likely granted virulence to *E. coli* K-12?

- A. Transformation [14%]
- B. Transduction [16%]
- ✔ C. Conjugation [63%]
- D. Transfection [5%]

Correct
62% answered correctly

Explanation:



Conjugation is the transfer of genetic information from one bacterial cell to another via direct contact. The donor cell contains the F (fertility) factor plasmid, a circular piece of DNA containing genes that direct the formation of the sex pilus. During conjugation, the sex pilus from the donor cell attaches to a recipient cell (one that does not contain F factor) and facilitates the transfer of a single strand of the F factor plasmid DNA to the recipient cell. The recipient then synthesizes a complementary strand and becomes capable of passing genetic information to another bacterium. The F factor plasmid is typically found outside the bacterium's genome, but it can integrate into the bacterial chromosome. When integration occurs, **bacterial genes** can be **transferred** with the F factor.

According to the question, *E. coli* K-12 is a nonvirulent bacterial strain. Nonetheless, co-culturing *E. coli* K-12 with **EHEC bacteria that contain F factor** resulted in *E. coli* K-12 acquiring virulence. *E. coli* K-12 virulence most likely resulted from the direct transfer of *LEE* genes through conjugation.

(Choice A) Transformation is the cellular uptake of foreign DNA directly from the environment. The *E. coli* K-12 bacteria in this scenario most likely received new DNA through interactions with nearby EHEC bacteria rather than directly from their surroundings.

(Choice B) Transduction involves DNA transfer from one bacterial cell to another by a bacteriophage (a virus that infects bacteria). During assembly of bacteriophages inside an infected cell, bacterial DNA can become trapped within the capsid of newly created bacteriophages. Subsequent infection of other cells with these new bacteriophages results in the transfer of bacterial DNA into a new host. Neither of the experimental *E. coli* cultures was exposed to bacteriophages.

(Choice D) Transfection is the process by which genetic material, usually in the form of a plasmid, is introduced into *eukaryotic* cells. *E. coli* K-12 is a *prokaryotic* cell, meaning that transfection does not apply to this scenario.

Educational objective:
Conjugation is the transfer of genetic information from one bacterial cell to another via direct contact facilitated by the sex pilus. The sex pilus is encoded by genes contained in the F factor plasmid.

Time spent: 0 sec
Subject: Biology

QID: 403595
System: Cellular Biology

Enterohemorrhagic *Escherichia coli* (EHEC) O157:H7 is a bacterium responsible for causing foodborne illnesses. EHEC virulence, or the ability to cause disease, is acquired through quorum sensing, a form of bacterial cell-cell communication dependent on population density. EHEC O157:H7 bacteria monitor population density by detecting changes in the concentration of autoinducer-3 (AI-3), an extracellular signaling molecule secreted by many bacterial species. AI-3 generally acts on neighboring bacteria to activate the genes on the enterocyte effacement (*LEE*) locus. These genes code for the type III secretion system (T3SS) as well as several secreted proteins (eg, EspA, EspB, EspD, Tir), which promote entry of Shiga-like toxins secreted by EHEC into host intestinal epithelial cells. This ultimately leads to cell death, lesions, and symptoms such as bloody diarrhea and abdominal cramps.

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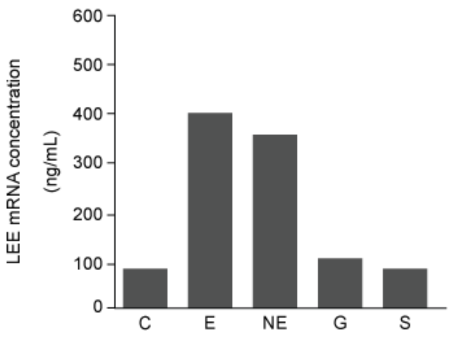


Figure 1 Concentration of *LEE* mRNA in mutant LuxS⁻ bacteria supplemented with purified peptide hormones (Note: C = control.)

A western blot was performed to compare the presence and concentration of type III secretion proteins from wild-type (WT), LuxS⁻, and LuxS⁻ cultured in preconditioned media (PCM). PCM was generated by culturing uninfected intestinal epithelial cells in growth media for 24 hours. After 24 hours, the cells were removed and the PCM was used to supplement the growth of LuxS⁻ bacteria.

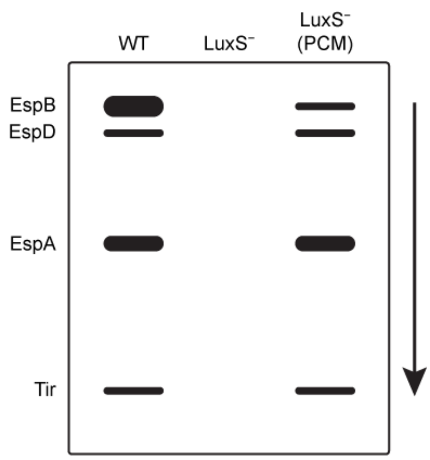


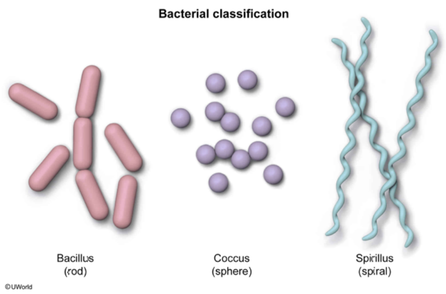
Figure 2 Western blot results of type III secretion proteins isolated from WT, mutant LuxS⁻, and LuxS⁻ in PCM

EHEC is classified as a bacillus bacterium. Given this information, this type of classification is most likely based on the bacterium's:

- A. habitat. [2%]
- B. oxygen dependence. [1%]
- ✔ C. morphology. [88%]
- D. virulence. [6%]

Correct
87% answered correctly

Explanation:



Prokaryotic organisms such as bacteria can be classified by their shape, or **morphology**. The morphology-based classification of bacteria involves three basic shapes:

- **Bacilli** are rod-shaped bacteria.
- **Cocci** are spherical bacteria.
- **Spirilli** are spiral bacteria.

This method of classification enables physicians to identify bacteria easily based on their appearance under a light microscope.

(Choice A) Habitat refers to the natural environment in which an organism resides. Classification according to habitat typically uses the suffix *–philes*. Bacteria tend to inhabit numerous environments; for example, some bacteria can be thermophiles (comparatively high temperatures), acidophiles (low pH), or halophiles (high salt).

(Choice B) Oxygen dependence is determined by a bacterium's ability to use fermentation, anaerobic respiration, aerobic respiration, or some combination of these to generate ATP from various nutrients.

(Choice D) Virulence refers to the degree to which a pathogen can be harmful, such that highly virulent pathogens can cause life-threatening conditions. Although bacteria can be classified based on their virulence, these classifications are not uniform.

Educational objective:
Bacteria can be classified by morphology (ie, shape). In morphological classification, three basic shapes are used to classify bacteria: rod-shaped (bacilli), spherical-shaped (cocci), and spiral (spirilli).

Time spent:0 sec
Subject:
Biology

QID:403596
System:
Cellular Biology

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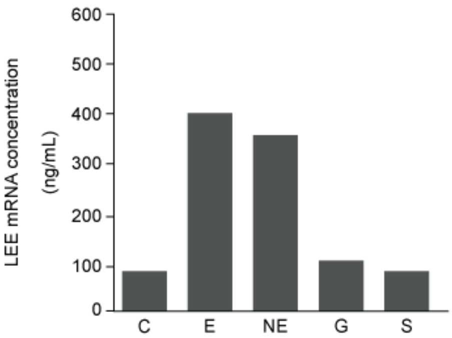

Figure 1 Concentration of *LEE* mRNA in mutant $LuxS^-$ bacteria supplemented with purified peptide hormones (Note: C = control.)

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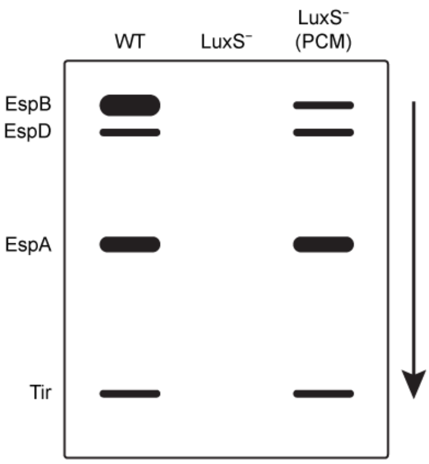


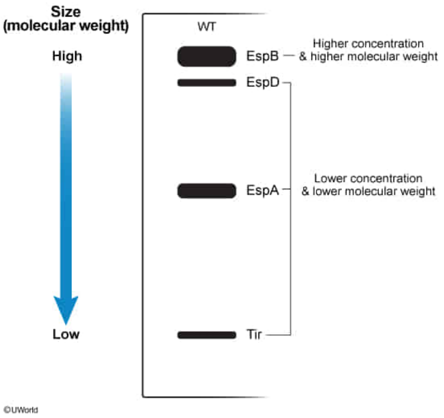
Figure 2 Western blot results of type III secretion proteins isolated from WT, mutant LuxS⁻, and LuxS⁻ in PCM

Which proteins identified in Figure 2 have the shortest amino acid sequence and the greatest concentration in WT cells, respectively? Assume all antibodies used detect their epitopes with equal effectiveness.

- A. EspA; Tir [4%]
- B. Tir; EspB [81%]
- C. EspB; EspA [9%]
- D. EspB; EspD [4%]

Correct
81% answered correctly

Explanation:



Western blot is a laboratory technique that first separates proteins by size (gel electrophoresis) and then transfers them to a blotting membrane. A primary antibody specific for the protein of interest is used as a probe. The membrane is washed to clear unbound antibodies and then treated with a secondary antibody that specifically binds to the primary antibody. Depending on the type of secondary antibody used, detection of the proteins of interest is achieved either by fluorescence or by a reporter enzyme. In general, proteins with the **shortest amino acid sequences** have lower molecular weights and **migrate quickly** through the gel. **Band intensity** (thickness) demonstrates the relative levels of **expression** of the target proteins.

In this experiment, EspA, EspB, EspD, and Tir were separated by gel electrophoresis according to their molecular weights. Figure 2 shows the relative size and expression levels of the type III secretion proteins isolated from WT, mutant LuxS⁻, and LuxS⁻ in PCM. The results demonstrate that **Tir** traveled farthest, indicating it had the **lowest molecular weight** and therefore the shortest amino acid sequence. The **EspB** protein had the **greatest concentration** in WT cells, as indicated by the **increased intensity** of its bands in Figure 2.

(Choice A) EspA has the second-lowest molecular weight, as it migrated less than Tir. Tir demonstrated less signal intensity than EspB.

(Choices C and D) EspB has the greatest molecular weight as it traveled the shortest distance compared with the other proteins. The EspD and EspA bands demonstrated less signal intensity compared with EspB bands.

Educational objective:
Western blot involves protein separation by gel electrophoresis, transfer onto a blotting membrane, and detection with protein-specific antibodies. Proteins with low molecular weight migrate farthest, and protein concentration is generally proportional to band intensity.

Time spent:0 sec
Subject:
Biology

QID:403597
System:
Cellular Biology

Enterohemorrhagic *Escherichia coli* (EHEC) O157:H7 is a bacterium responsible for causing foodborne illnesses. EHEC virulence, or the ability to cause disease, is acquired through quorum sensing, a form of bacterial cell-cell communication dependent on population density. EHEC O157:H7 bacteria monitor population density by detecting changes in the concentration of autoinducer-3 (AI-3), an extracellular signaling molecule secreted by many bacterial species. AI-3 generally acts on neighboring bacteria to activate the genes on the enterocyte effacement (*LEE*) locus. These genes code for the type III secretion system (T3SS) as well as several secreted proteins (eg, EspA, EspB, EspD, Tir), which promote entry of Shiga-like toxins secreted by EHEC into host intestinal epithelial cells. This ultimately leads to cell death, lesions, and symptoms such as bloody diarrhea and abdominal cramps.

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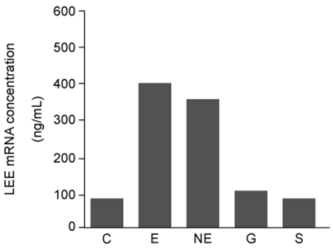

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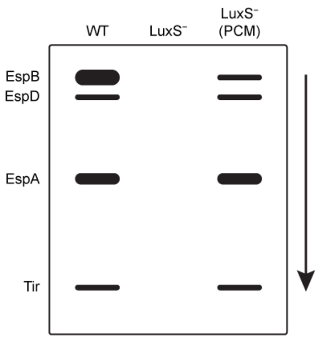


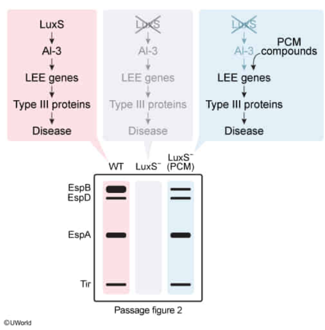
Figure 2 Western blot results of type III secretion proteins isolated from WT, mutant LuxS⁻, and LuxS⁻ in PCM

Based on the passage, what best explains the western blot results of type III secretion proteins in LuxS⁻ *E. coli* cells cultured in PCM?

- A. Intestinal epithelial cells secreted secretin and gastrin into the PCM. [7%]
- B. Type III secretion proteins were produced by intestinal epithelial cells. [14%]
- C. LuxS⁻ cells underwent a spontaneous reverse mutation that restored normal LuxS function. [10%]
- ✔ D. Compounds released by intestinal epithelial cells into the PCM activated transcription of *LEE* genes. [67%]

Correct
67% answered correctly

Explanation:



In quorum sensing, changes in gene expression are dependent on the cellular density of bacteria that produce and secrete the autoinducer, a signaling molecule that influences the activity of specific genes in nearby cells (AI-3, in this case). The amount of autoinducer in extracellular space correlates with bacterial population density.

According to the passage, deficiency in the LuxS enzyme leads to decreased AI-3 expression. AI-3 generally acts on neighboring bacteria to induce expression of *LEE* genes. These genes encode the type III secretion proteins necessary to introduce EHEC-secreted toxins into the host cells to cause disease. Because LuxS⁻ mutants lack the LuxS enzyme and have insufficient AI-3 levels, these mutants lack type III secretion proteins, as shown in Figure 2.

However, Figure 2 also shows that growing LuxS⁻ cells in PCM resulted in the expression of type III secretion proteins. Based on Figure 1, which shows that

LEE genes can be expressed via a **LuxS-independent method**, the banding patterns in Figure 2 suggest that **compounds** (eg, epinephrine, norepinephrine) released by epithelial cells **into the PCM activated the transcription** of *LEE* genes despite the deficiency of AI-3.

(Choice A) Figure 1 shows that secretin and gastrin affect the concentration of LEE mRNA at a level that is comparable to the control group, LuxS⁻ (ie, no significant change compared to the control). As a result, it would not be expected that these hormones rescue the production of type III secretion proteins in LuxS⁻ cells.

(Choice B) The passage states that type III secretion proteins can be secreted only by bacteria, not by the intestinal epithelial cells.

(Choice C) A reverse mutation for LuxS⁻ *E. coli* would result in the western blot results showing similar banding patterns for all three *E. coli* cultures. However, this is not the case as LuxS⁻ *E. coli* show a reduced banding pattern compared with the WT.

Educational objective:
Inducers are chemical substances (signals) that regulate gene expression. Specifically, secreted inducers can influence gene expression in other cells.

Time spent:0 sec
Subject:
Biology

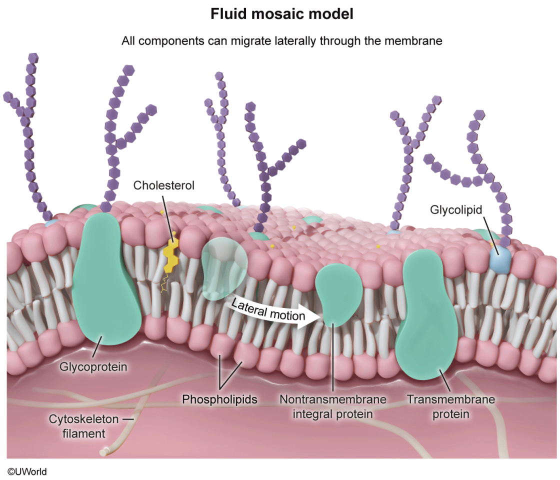
QID:403598
System:
Cellular Biology

Which of the following molecules is most likely found within the cell membranes of intestinal epithelial cells?

- A. Peptidoglycan [16%]
- B. Cholesterol [59%]
- C. Cytoskeletal filaments [20%]
- D. Cellulose [2%]

Correct
58% answered correctly

Explanation:



The cell membrane is a **phospholipid bilayer** that separates the interior of the cell (ie, intracellular space) from the external environment (ie, extracellular space). The two phospholipid layers are oriented such that the hydrophobic tails of the phospholipids in the inner and outer layers are in contact with each other, leaving the hydrophilic head groups in contact with either the intracellular or the extracellular fluid environment. The cell membrane is described by the **fluid mosaic model** as a nonrigid (fluid) phospholipid bilayer through which diverse proteins are scattered (the mosaic) and are able to *migrate laterally*.

In addition to phospholipids, animal cell plasma membranes also contain cholesterol, glycolipids (ie, lipids with attached sugar groups), glycoproteins (ie, proteins with attached sugar groups), and other proteins. **Cholesterol**, which is found in eukaryotic but not prokaryotic cell membranes, functions to regulate the fluidity of the bilayer. Due to their unique structure, cholesterol molecules decrease fluidity at higher temperatures and increase fluidity at lower temperatures. As **eukaryotic** cells, intestinal epithelial **cell membranes contain cholesterol**.

(Choice A) All cells, eukaryotic and prokaryotic, have a plasma membrane but some cell types are surrounded by an additional enveloping layer known as the cell wall. Unlike the plasma membrane, the cell wall is a *rigid* structure that defines the shape of the cell and provides added protection. Animal cells, including intestinal epithelial cells, *do not have cell walls*; however, nearly all prokaryotic cells have a cell wall. The major structural component of the bacterial cell wall is *peptidoglycan*, a polymer of sugars and short polypeptides.

(Choice C) Cytoskeletal filaments are found in the *cytoplasm* of cells and can interact directly with various components of the cell membrane (eg, by anchoring membrane proteins in place). However, they are not themselves part of the plasma membrane.

(Choice D) Cellulose, a polysaccharide, is the primary component of the cell wall in plant cells. Cellulose is *not* found in the membranes of animal cells.

Educational objective:
The plasma membrane is a fluid lipid bilayer that contains a diverse array of proteins scattered throughout. Certain cell types have a cell wall surrounding their plasma membrane; however, animal cells do not have a cell wall. The primary component of bacterial cell walls is peptidoglycan.

Time spent:0 sec
Subject:
Biology

QID:403599
System:
Cellular Biology